

LAHIRUNI MALSHIKA

Software Engineer

 Matara, Sri Lanka
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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building full-stack, AI-integrated web and mobile applications across the MERN stack and Flutter. Skilled in backend services, RESTful APIs, and responsive, accessible UIs blended with real-time data. Committed to clean, well-tested, maintainable code, code reviews, and software engineering best practices. An active, independent learner, eager to pick up new languages, frameworks, and tools and adapt to them quickly as team needs evolve.

WORK EXPERIENCE

Trainee Software Engineer at Virstack Pvt Ltd

DEC 2025 - AUG 2026

Tech Stack : MERN, Flutter/Dart, Docker, AWS, NGINX, CI/CD, WebSockets

- Contributed to developing and maintaining AI-powered full-stack applications using the MERN stack, as well as cross-platform mobile development with Flutter.
- Optimized application performance for fast, reliable user experiences across web and mobile, integrated AI-driven features to enhance product functionality, and built scalable interfaces with clean software architecture.
- Supported DevOps workflows through Docker containerization and AWS cloud infrastructure, collaborating within an agile team using version control to deliver production-ready solutions.

Intern Software Engineer at Virstack Pvt Ltd

MAR 2025 - NOV 2025

Tech Stack : MERN, Angular, Flutter/Dart

- Contributed to full-stack web and mobile application development using React, Angular, Express with TypeScript, and Flutter (Dart).
- Optimized RESTful APIs and MongoDB aggregation pipelines for search, filtering, scheduling, and data-driven workflows, improving data retrieval efficiency and application responsiveness.

EDUCATION

BSc (Hons) in Information Technology - University of Moratuwa

JUN 2022 - JUL 2026

- Dean's List: Level 2 Semester 1 - SGPA: 3.89
- Dean's List: Level 2 Semester 2 - SGPA: 3.94
- Dean's List: Level 4 Semester 1 - SGPA: 3.85
- CGPA: 3.48 (out of 4.0 - up to L4S1)

Diploma in English - BritishWay English Academy

FEB 2021 - MAR 2021

G.C.E. Advanced Level - Sujatha Vidyalaya Matara

JUN 2017 - OCT 2020

- Stream - Physical Science

TECHNICAL SKILLS

Programming Languages:	Java, C, Python, JavaScript, TypeScript, Dart
Frontend Development:	React JS, Next JS, JavaScript, Tailwind CSS, HTML, CSS, Flutter
Backend Development:	Node JS, Express JS, Nest JS
Databases:	MySQL, MongoDB, Neo4j
Cloud & Deployment:	Docker, Kubernetes, CI/CD Pipelines, AWS S3/EC2/CloudWatch, NGINX
Testing & Quality Assurance:	Cypress, Selenium, Unit Testing, End-to-End Testing
Software Engineering Principles:	Git, OOP, Design Patterns, SOLID principles, REST APIs, Microservices Architecture, Agile/Scrum, Secure Software Engineering

SOFT SKILLS

- Problem Solving
- Time Management
- Leadership
- Communication
- Teamwork & collaboration

PROJECTS

Final Year Research - Exploring Hardware-Software Co-design of Spatial Deep Learning Hardware

Accelerators (Group project)

Developed a cycle-accurate hardware-software co-design simulator for systolic-array DNN accelerators to address performance bottlenecks in DNN execution. Implemented a SystemVerilog RTL model with a Python compiler/mapper for memory management in the simulator.

- Technologies: Python, Verilog, DNN Workloads, FPGA Simulation
- GitHub Repository

Fuel Price Prediction Dashboard using Machine Learning (Individual Project)

Built a machine learning model to predict commonly used fuel prices in Sri Lanka. Performed data preprocessing, feature engineering, and classification.

- Technologies: Python, Machine Learning, Pandas, Scikit-learn
- GitHub Repository

Travel Recommendation Expert System (Individual project)

A rule-based expert system recommending travel destinations via logical inference, with a Prolog knowledge base blended into a Python GUI; features include best-season prediction and cheapest-destination finding.

- Technologies: Python, Prolog (SWI-Prolog), Tkinter
- GitHub Repository

MozzaMelt - Ecommerce Platform (Group Project)

A web application for a pizza shop supporting ordering, payments, order tracking, profile management, and reviews. Contributed to front-end development and quality assurance, including the CI/CD pipeline and automated testing.

- Role - Frontend Developer, Software Quality Assurance Engineer
- Technologies - React JS, ASP.NET Core, MS SQL, Docker, CI/CD Pipeline, Selenium
- Frontend GitHub Repository • Backend GitHub Repository

Anothershots - Photography Focused Web Application (Level 2 Software Project - Mentored by 99X)

An inclusive platform connecting photography professionals with clients. Implemented the Featured Photo, Contact, and Packages sections of the photographer profile, plus the Booking and Admin Dashboard, applying clean architecture and design patterns.

- Role - Full Stack Developer
- Technologies - Next JS, TypeScript, Tailwind CSS, Nest JS, Prisma ORM, MongoDB
- Frontend GitHub Repository • Backend GitHub Repository

ACHIEVEMENTS AND PARTICIPATIONS

- **CyberZee'24 - Top 06** : An inter university quiz competition arranged by University of Kelaniya.
- **ACES Coders V11.0 2024 - Participant** : An inter university coding competition organized by University of Peradeniya.
- **MoraXtreme8.0 2023 - Top 75** : An inter university coding competition coordinated by IEEE student branch of University of Moratuwa.
- **Code Rush 2023 - Intra Faculty Coding Competition** : Participation
- **RealHack 5.0 - Participant** : An inter university coding competition organized by University of Kelaniya.

EXTRACURRICULAR ACTIVITIES

- **IEEE Student Branch of UoM** : Publicity committee member - Rise Up Mora 2023
- **IEEE Power & Energy Society Chapter of UoM** : WebMaster (Term of 24/25)
- **IEEE Women In Engineering Affinity Group of UoM** : Chief Organizer (Term 24/25), Marketing Committee Lead
- **Mathematics Society of UoM** : Publicity committee member - ENIGMA 2024
- **Leo Club of UoM**

REFERENCES

Dr. T.M. Thanthriwatta

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